WeIntern Data Science Internship — Week 1

E-Commerce Data Cleaning, Sales Analysis & Visualization

Dataset note: The WeIntern handbook did not include a dataset. A synthetic e-commerce dataset was therefore created to match the requested fields and demonstrate the required cleaning workflow transparently.

1. Objective

Build disciplined habits in data cleaning, KPI analysis, trend identification, visualization, business interpretation, and professional reporting.

2. Dataset Overview

The raw dataset contains 503 rows and 11 columns. It includes order, customer, product, quantity, price, date, payment, delivery, and city fields.

3. Cleaning Summary

Check	Result
Initial rows	503
Final rows	499
Missing quantity handled	3
Missing city handled	8
Missing payment mode handled	5
Exact duplicates removed	3
Invalid dates handled	1
Negative quantity corrected	3
Non-positive prices corrected	2

4. KPI Summary

KPI	Value
Total Revenue	8683007.44
Total Orders	367
Average Order Value	23659.42
Total Units Sold	1075
Top-Selling Category	Electronics
Top Revenue-Generating Product	Laptop

5. Trend and Customer Insights

• 2025-06 recorded the highest completed-sales revenue at 1,619,256.61.

• 2025-01 recorded the lowest completed-sales revenue at 1,253,978.02.

• Electronics is the top revenue-generating category, contributing approximately 68.4% of completed-sales revenue.

• Laptop is the top revenue-generating product in the cleaned completed-sales dataset.

• Jaipur has the highest number of completed orders with 59 orders.

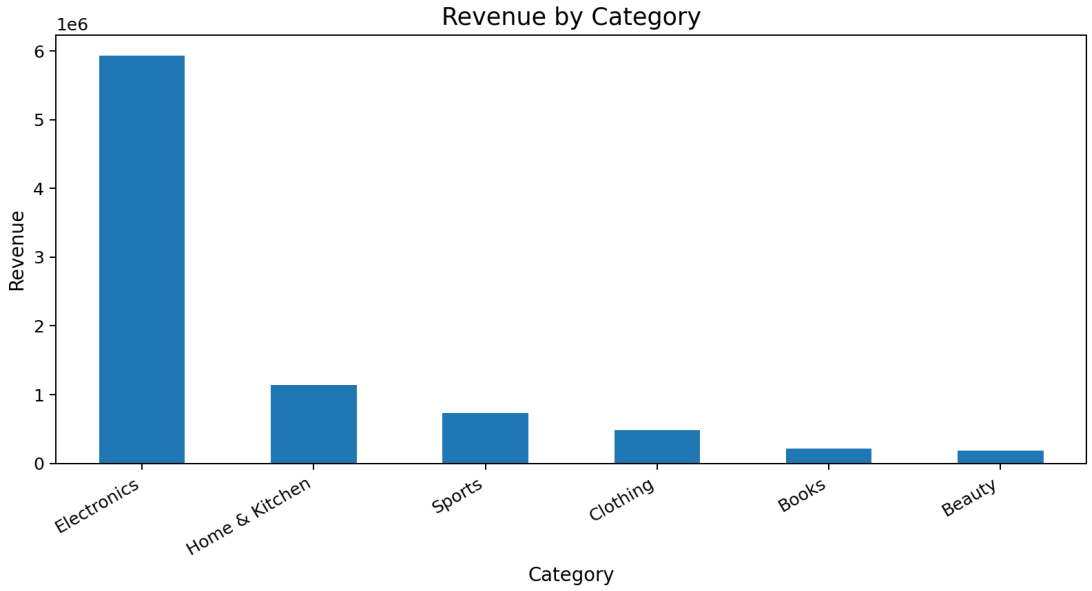
• The average completed order value is 23,659.42, providing a benchmark for typical customer order size.

• 1075 units were sold across 367 completed orders in the analysis period.

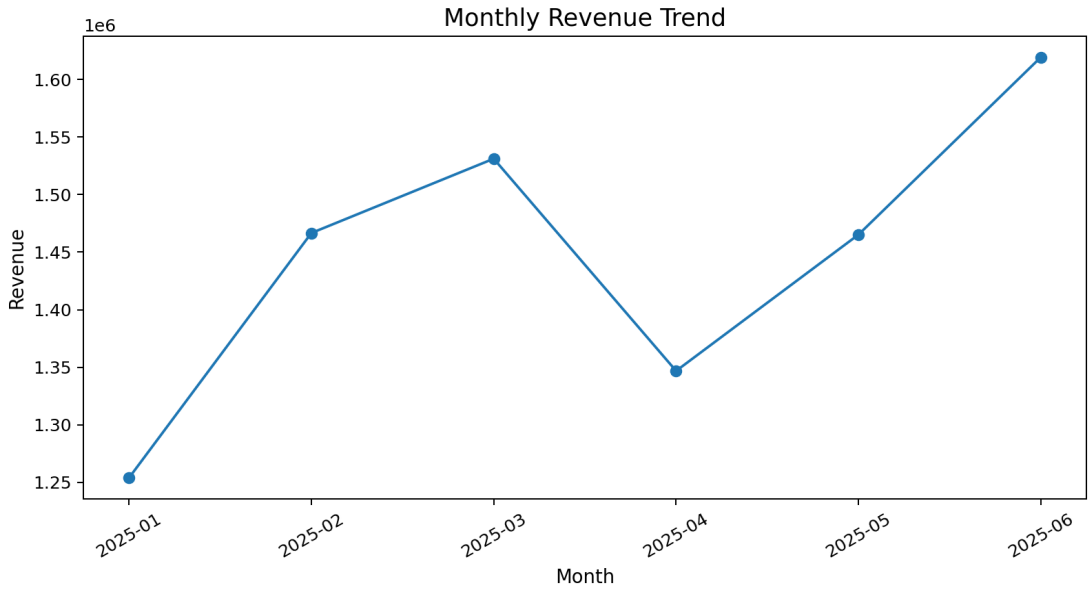
• Debit Card contributes the largest share of completed-sales revenue among payment modes.

6. Visualizations

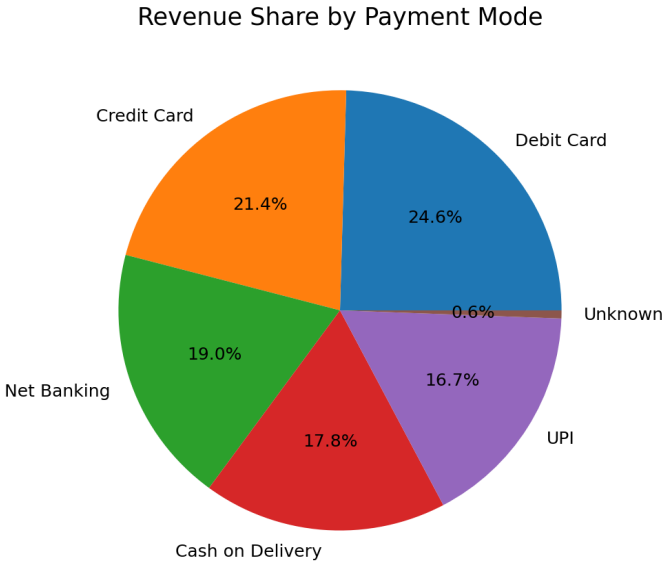
Revenue by Category



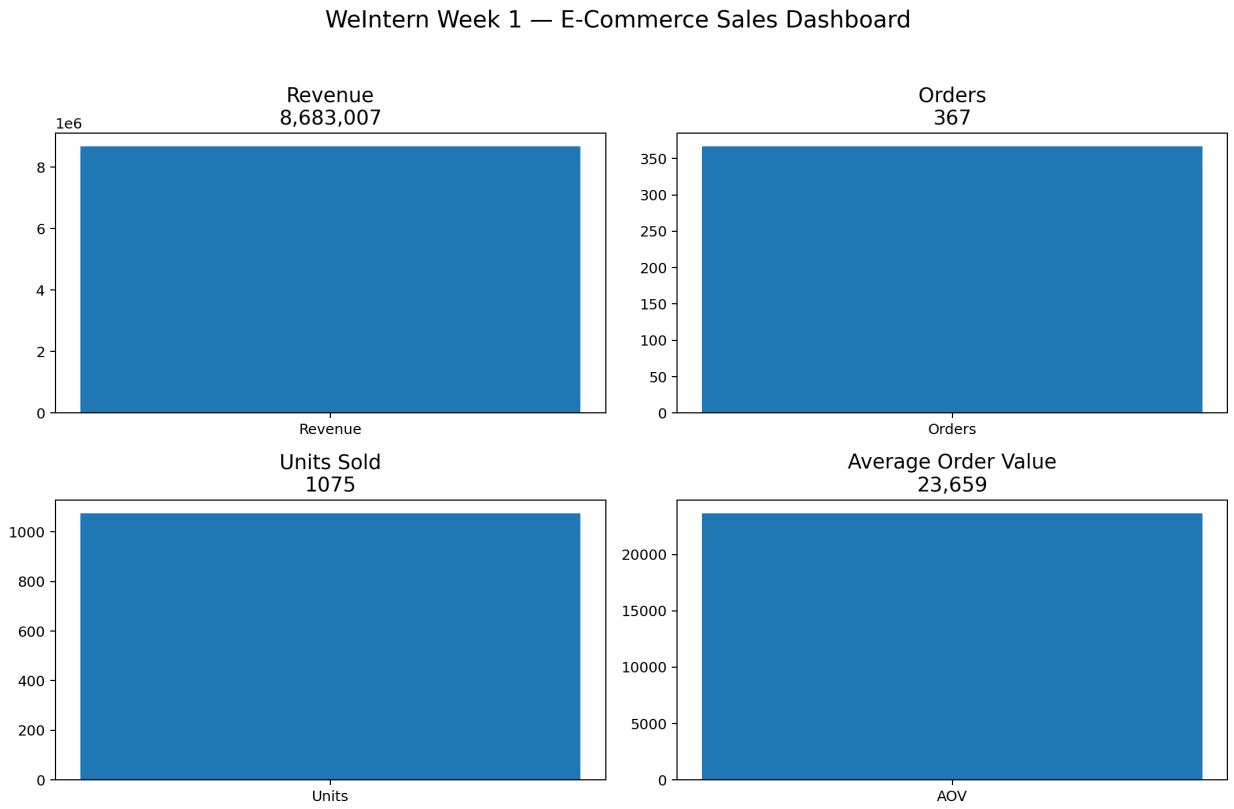
Monthly Revenue Trend



Revenue Share by Payment Mode


Revenue Share by Payment Mode
Credit Card
Debit Card
21.4%
24.6%
0.6%
Unknown
19.0%
Net Banking
16.7%
UPI
17.8%
Cash on Delivery

7. Dashboard Mockup



8. Recommendations

• Prioritize strong revenue categories in promotions while reviewing weaker categories.

• Study the highest-performing month for repeatable campaign or seasonal factors.

• Use city-level order concentration to guide local marketing and delivery capacity.

• Develop retention offers for repeat and high-value customers.

• Keep the strongest payment methods convenient for customers.

9. Limitations and Assumptions

The dataset is synthetic because no dataset was supplied with the WeIntern handbook. Completed sales are defined as records with Delivery Status = Delivered. Recommendations are based only on this dataset and should be validated before use in a real business.